Harith Al-Safi

Full Stack Engineer

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About me

A Full Stack Engineer with **6 years** of experience specializing in **Java Spring Boot**, React.js, Machine Learning and Cloud Technologies. Proven record in leading technical teams and delivering **GraphQL** and **LLM-powered** platforms across **startup** to **enterprise**.

Experience

Full Stack Engineer
Sep 2024 - Ongoing (2+ Years)
BT Group
Manchester, UK

- Owned a **GraphQL subgraph** on BT's **Apollo Federation Supergraph** end to end, from discovery to onboarding
- Ran a **consulting phase** with frontend and backend teams to unify multiple services behind **one GraphQL API**
- Demoed the subgraph to **400+ engineers**, cutting frontend round trips and over-fetching across services
- Hardened **Spring Boot** services with **caching** and **circuit breakers**, deployed to **Kubernetes** via **Helm**
- Orchestrated **multiple AI coding agents** to deliver the subgraph fast with robust, high-quality code
- Delivered 5 **Kafka-based Spring Boot microservices** with MongoDB/PostgreSQL, Netflix Conductor, and **Pulumi**
- Achieved **30% AWS cost reduction** by automating infrastructure with TypeScript and optimizing **ECS cluster**
- Re-architected Spring Boot APIs with JWT, **Azure AD**, and **role-based access** achieving **enterprise grade security**
- Led **React.js to Next.js** migration of 5K+ lines, adding Cypress testing and OpenAPI to TypeScript **code gen**
- Mentored **5+ apprentices** in **Java 21** Spring Boot and ran the regional Graduate Network's monthly tech talks

Technical Lead (Part-time)
Oct 2025 - Ongoing (1 Year)
Freelance
Remote

- Directed a team building a full-stack **AI document search platform** hosted on **Google Cloud (GCP)**
- Engineered a robust text-based **search engine with Elasticsearch** for fast, relevant document retrieval
- Evaluated **enterprise-grade LLMs** from **Gemma** to **Gemini** under enterprise data-protection requirements
- Tested multiple **document processing pipelines** to select the most accurate approach for LLM-based extraction
- Designed a **background queueing system** for LLM calls, turning **1–2 minute waits into instant responses** for users

Lead Frontend Developer
Aug 2023 - Nov 2023 (4 Months)
Velz Travel
Remote

- Directed the frontend of a travel web app for a startup built with **Redux Toolkit, Next.js** and **Material UI**
- Achieved 93% test coverage through comprehensive unit testing, resulting in **70% reduction in production bugs**

Software Engineer
July 2022 - July 2023 (12 Months)
Johnson Controls
London, UK

- Invented a **scripting language with a compiler** for fire detection, increasing development productivity by 40%
- Managed a **team of 10** interns to create a **business plan** and ended up placing 2nd amongst the whole company
- Administered C/C++ projects while fixing bugs, adding features, refactoring and **migrating 90% of codebase to C#**
- Collaborated on an **analytics web dashboard with React.js** for predictive maintenance with dev ops and CI/CD

Full Stack Developer
Dec 2021 - May 2022 (6 Months)
Paperound - Freelance
Leeds, UK

- Developed a scalable analytics dashboard in Next.js, with **MSAL** and **Auth0** for seamless, secure authentication
- Built and trained a **reinforcement neural network** using **TensorFlow**, KerasRL for predicting market movements
- Developed **regression models** with Python, **Scikit-learn** and Matplotlib, and presented to business **stakeholders**

Information Technology Administrator
Jun 2020 - Jun 2021 (12 Months)
Mesopotamia Group
Amman, Jordan

- Developed **high-availability load balancer with C/MPI** achieving 34% improved system uptime
- Built **predictive analytics pipeline using Python** for proactive cybersecurity threat detection

Education

BEng (Honours) in Electronics and Computer — Sept 2019 - July 2024
University of Leeds (UoL) — First-Class Honours (4.0/4.0 GPA)

- Cloud and Parallel computing, Secure and Distributed systems, Cybersecurity, Software Engineering, Data Science
- Embedded systems, Microprocessors, FPGA, Circuit Analysis, Signal processing, Networking, Compiler design

Skills and Languages

Backend: Java 21 | Spring Boot | GraphQL | Apollo Federation | Kafka | Conductor | Python | C#

Frontend: TypeScript | React.js | Next.js | Redux | Tailwind CSS | Material UI | Cypress | Express.js

ML / AI: LLMs (Gemini, Gemma) | RAG | Elasticsearch | TensorFlow | Scikit-learn | XGBoost | Pandas

Cloud: GCP | AWS (Lambda, ECS, RDS, S3) | Azure (AD, Serverless) | Kubernetes | Helm | Docker | Pulumi

Data / Tools: PostgreSQL | MongoDB | Redis | Firestore | SQLite | Weaviate | Jenkins | Git | Jira

Soft: Technical Leadership | Mentoring | Stakeholder Management | Communication | Problem solver

Languages: **Arabic** (Full Professional) **English** (Full Professional) **Spanish** (Elementary)

Publications

Vega: LLM-driven Intelligent Chatbot Platform for Internet of Things Control and Development
2025
MDPI Sensors - 10.3390/s25123809 - 1st Author

- Developed a full stack **Chatbot** web application with TypeScript, React.js, Next.js, **TailwindCSS** and **Redis**
- Programmed language processing algorithms using **multi-modal large language models** to execute user messages
- Implemented a **Python Flask** REST Server with **SQLite** on the **Raspberry Pi** to communicate with the chatbot
- Engineered a **circuit** with sensors, camera, fans, LCD and buttons to execute user's commands from the chatbot

Projects

CV Studio — Oct 2024 - Ongoing — Personal - Beta on cvstudio.ai

- Built a **full-stack** resume-improvement web app powered by LLMs, serving **hundreds of active users**
- Implemented **live resume editing** with ShadcnUI and Typst **WebAssembly** PDF compilation for real-time render
- Integrated **Clerk authentication** and **Stripe subscription** payments for secure user access and monetization
- Architected a RAG pipeline with **Weaviate vector DB** and **OpenAI embeddings** for resume analysis

Green ML: Machine Learning for Green Bond Syndication — May 2024 - Aug 2024 — Personal

- Engineered bond syndication **prediction model** using **XGBoost** and network analysis on 3K+ institutional holders
- Implemented **parallel processing** with Ray to extract network features from adjacency matrices with **70% speedup**
- Built machine learning pipeline achieving **0.95+ AUC-ROC** for predicting bond syndication collaboration patterns
- Processed **Bloomberg financial data** using **pandas**, **NetworkX** for multi-regional investment analysis

MCPP: High Performance C++ Library for Numerical Computation — Apr 2021 - Oct 2022 — Personal

- Created a **cross-platform math library** for C++ used for matrix algebra, statistics and machine learning
- Engineered with **CI/CD** using **Jenkins**, unit tests, Docker build, **Cmake** and Doxygen documentation
- Improved performance by 40% using **parallel computing** with **OpenMP**, OpenCL and **CUDA**